Electrospun PVDF Nanofiber Membrane

Product Description

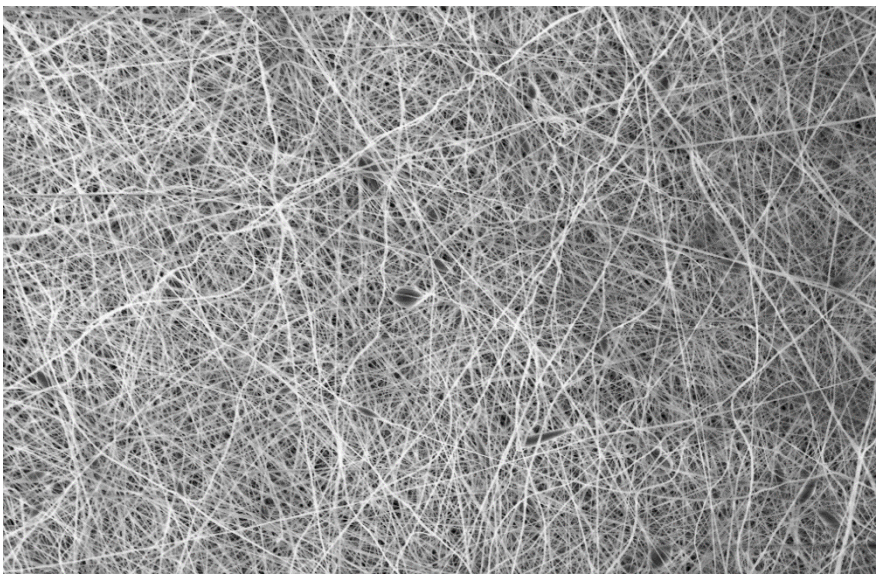
E-Spin Nanotech manufactures high-performance electrospun PVDF nanofiber membranes designed for advanced filtration, sensing, and energy storage applications. Polyvinylidene difluoride (PVDF) is a chemically stable thermoplastic fluoropolymer known for its excellent mechanical strength, thermal stability, and strong piezoelectric properties. It is lightweight, flexible, chemically inert, and highly durable under demanding operating conditions, making it an ideal material for functional nanofiber membranes.

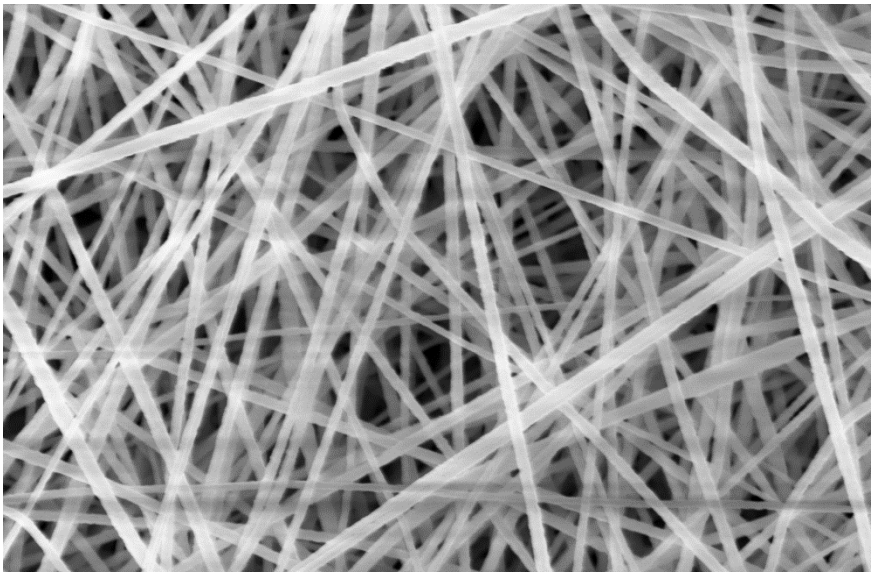
Electrospun PVDF membranes possess high porosity, ultrafine fibre diameter, and strong electrostatic properties, which enable efficient capture of fine particulate matter in air filtration systems. Their controlled pore structure and chemical resistance also make them suitable for gas separation and advanced filtration processes.

Due to their intrinsic piezoelectric behaviour, PVDF nanofibers are widely used in sensor technologies, where they convert mechanical stimuli such as pressure, vibration, or motion into electrical signals. Additionally, PVDF nanofiber membranes are widely used in energy storage systems, particularly as lithium-ion battery separators, where they provide excellent electrolyte compatibility, thermal stability, and mechanical integrity.

Key Applications

- Air filtration membranes
- Gas separation systems
- Sensors and piezoelectric devices
- Semiconductor processing
- Battery separators for energy storage systems





Membrane Specifications

Nanofiber Spinning Capability

- Pure PVDF
- PVDF blended with other polymers, drugs, nanoparticles etc.

Membrane Construction

- Free-standing nanofiber membrane
- Supported on substrates (e.g., aluminium foil, non-woven PP fabric, etc.)

Nanofiber Diameter Range

- 100 – 300 nm

Thickness and weight Density

- Thickness 60-100 µm
- Up to 8.0 gsm

Custom Sizes

- Rolls
- Sheets (10 × 10 cm, 14 × 28 cm, 25 × 200 cm)
- Custom-sized patches (as per specific requirements)

Polysulfone Nanofiber Membrane

Product Description

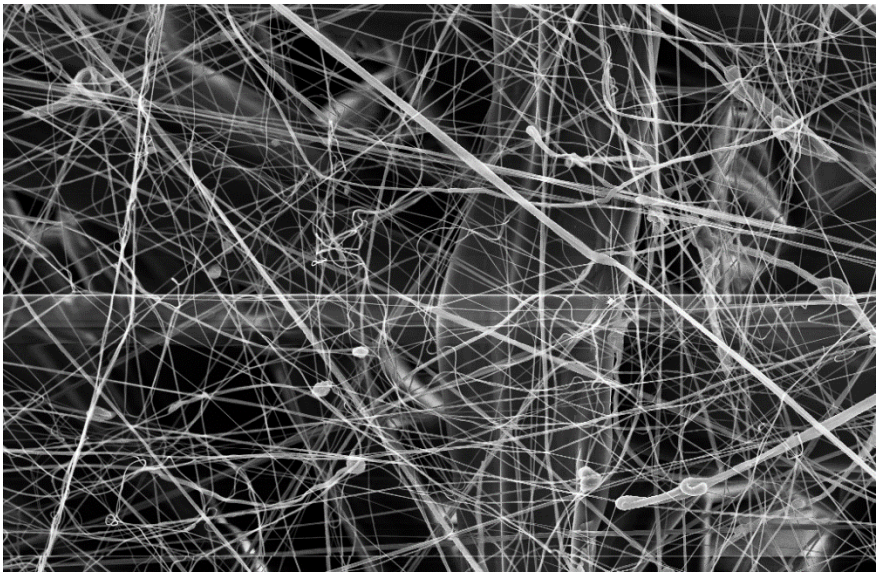
E-Spin Nanotech manufactures high-performance electrospun Polysulfone nanofiber membranes designed for advanced energy and filtration applications. These membranes feature a highly porous nanofibrous structure, excellent thermal stability, and superior chemical resistance, making them ideal for use in battery separators and other high-efficiency systems.

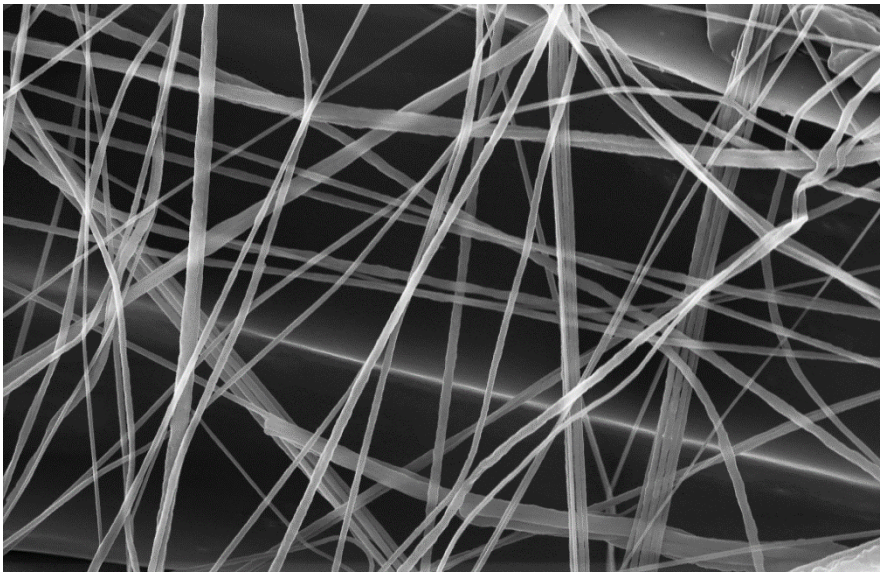
Our Polysulfone nanofiber membranes are engineered to deliver enhanced electrolyte uptake, improved ionic conductivity, and reliable mechanical strength, which are critical for next-generation battery technologies.

In addition to energy storage systems, these membranes are also suitable for a wide range of applications, including air filtration, water purification, and sensor technologies, where high surface area and controlled porosity are essential for performance.

Key Applications

- Battery separators for energy storage systems
- Air filtration membranes
- Water filtration and purification
- Sensor and advanced functional materials





Nanofiber Spinning Capability

- Pure Polysulfone
- Blended with other polymers, nanoparticles, additives etc.

Membrane Construction

- Free-standing nanofiber membrane

- Supported on substrates (e.g., aluminium foil, non-woven PP fabric, etc.)

Nanofiber Diameter Range

- 200 – 500 nm

Thickness and weight Density

- Thickness 50-80 µm
- Up to 4.0 gsm

Custom Sizes

- Rolls
- Sheets (10 × 10 cm, 14 × 28 cm, 25 × 200 cm)
- Custom-sized patches (as per specific requirements)

Nylon 6 Nanofiber Membrane

Product Description

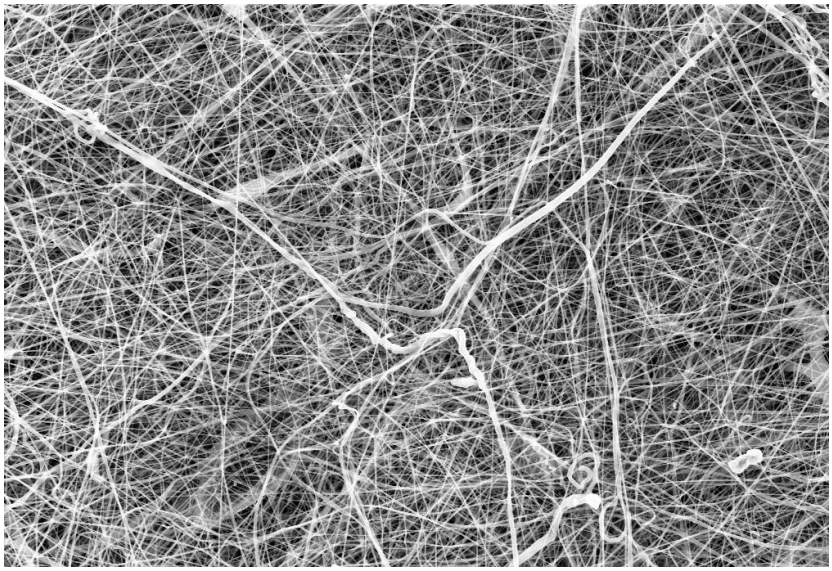
E-Spin Nanotech manufactures high-quality electrospun Nylon 6 nanofiber membranes designed for advanced filtration and functional material applications. Nylon 6 (polyamide 6) is a widely used engineering polymer known for its excellent mechanical strength, abrasion resistance, chemical stability, and good thermal properties, making it suitable for durable nanofiber membrane systems.

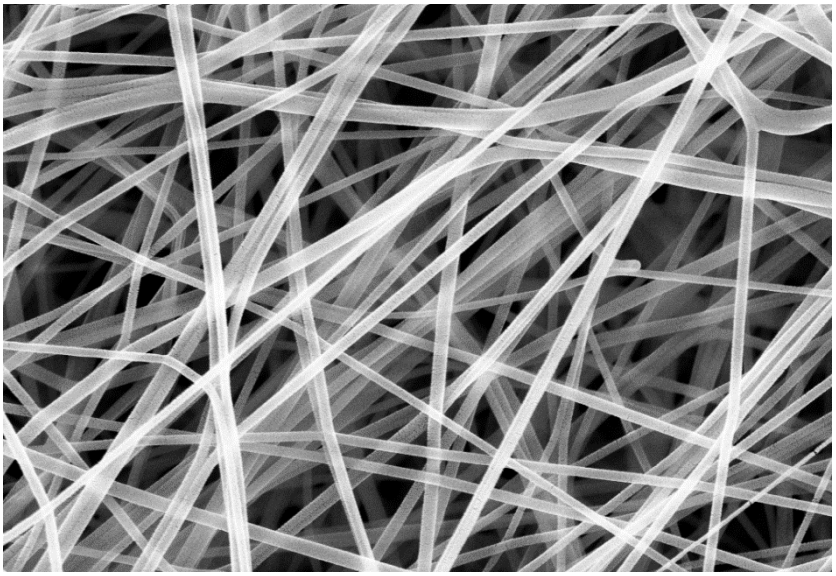
Electrospun Nylon 6 nanofiber membranes possess high porosity, interconnected pore structure, and ultrafine fibre diameters, which provide excellent permeability and high surface area. These characteristics enable efficient capture of fine particles, making them highly effective for air filtration and liquid filtration applications.

Due to their strong mechanical properties and chemical resistance, Nylon 6 nanofiber membranes are also suitable for nanofiber coating, protective textiles, and advanced separation technologies. Their flexibility and durability allow them to perform reliably in demanding environmental and industrial conditions**.**

Key Applications

- Air filtration membranes
- Protective and functional textiles
- Industrial separation processes
- Advanced filtration systems





Nanofiber Spinning Capability

- Pure PA6/PA66
- Blended with other polymers, nanoparticles, additives etc.

Membrane Construction

- Free-standing nanofiber membrane
- Supported on substrates (e.g., aluminium foil, non-woven PP fabric, etc.)

Nanofiber Diameter Range

- 70-100 nm

Thickness and weight Density

- Thickness 50-120 µm
- Up to 8.0 gsm

Custom Sizes

- Rolls
- Sheets (10 × 10 cm, 14 × 28 cm, 25 × 200 cm)
- Custom-sized patches (as per specific requirements)

PAN nanofiber membrane

Product Description

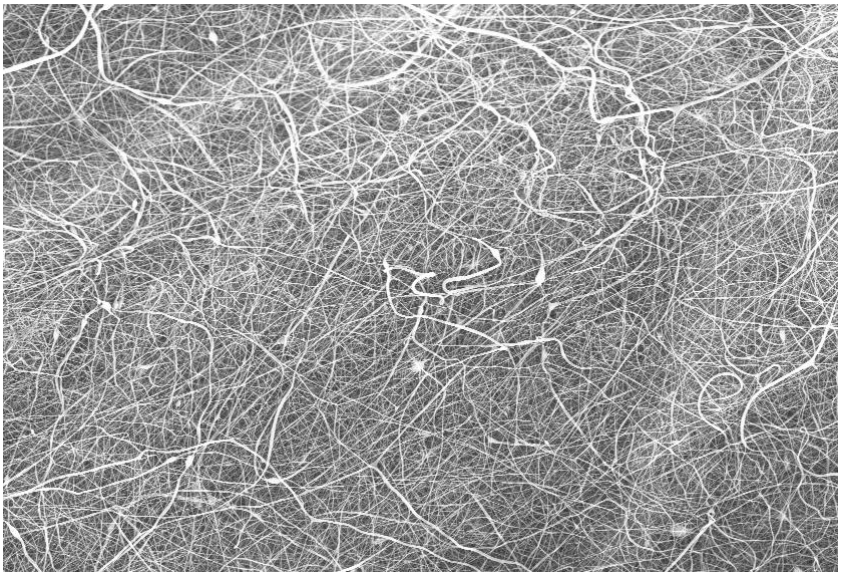
E-Spin Nanotech manufactures high-quality electrospun PAN (Polyacrylonitrile) nanofiber membranes designed for advanced filtration, energy, and functional material applications. Polyacrylonitrile is a widely used polymer known for its excellent mechanical strength, chemical resistance, and thermal stability, making it a reliable material for nanofiber membrane fabrication.

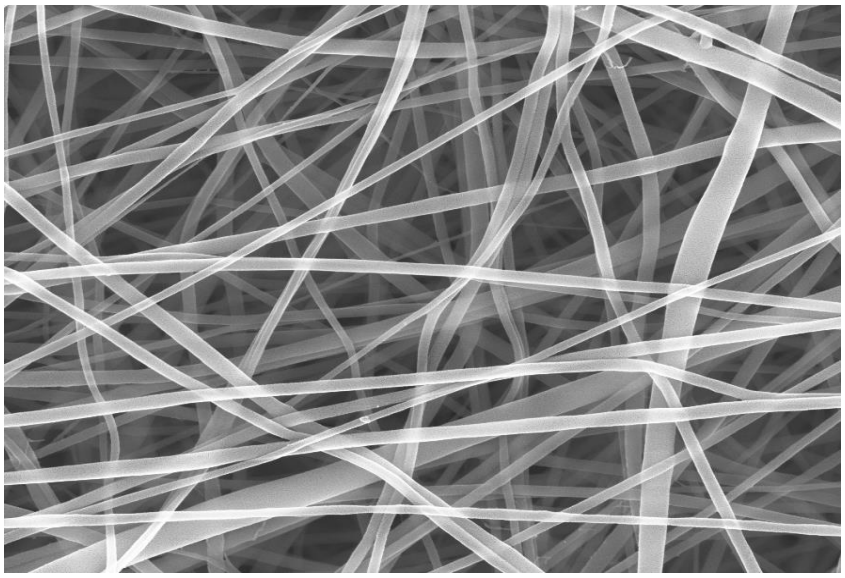
Electrospun PAN nanofiber membranes exhibit high porosity, interconnected pore structure, and ultrafine fiber diameter, which provide a large surface area and excellent permeability. These characteristics make them highly effective for air filtration, water purification, and fine particle removal.

PAN nanofiber membranes also serve as an important precursor material for carbon nanofibers, making them valuable in energy storage devices, electrodes, and advanced composite materials. Their stable structure and tunable properties allow them to be tailored for specific industrial and research applications.

Key Applications

- Air filtration membranes
- Water filtration and purification
- Energy storage materials and battery components
- Carbon nanofiber precursor materials
- Sensors and advanced functional materials





Nanofiber Spinning Capability

- Pure PAN powder
- Blended with other polymers, nanoparticles, additives etc.

Membrane Construction

- Free-standing nanofiber membrane
- Supported on substrates (e.g., aluminium foil, non-woven PP fabric, etc.)

Nanofiber Diameter Range

- 100-500 nm

Thickness and weight Density

- Thickness 50-100 µm
- Up to 6.0 gsm

Custom Sizes

- Rolls
- Sheets (10 × 10 cm, 14 × 28 cm, 25 × 200 cm)
- Custom-sized patches (as per specific requirements)

PVA nanofiber membrane

Product Description

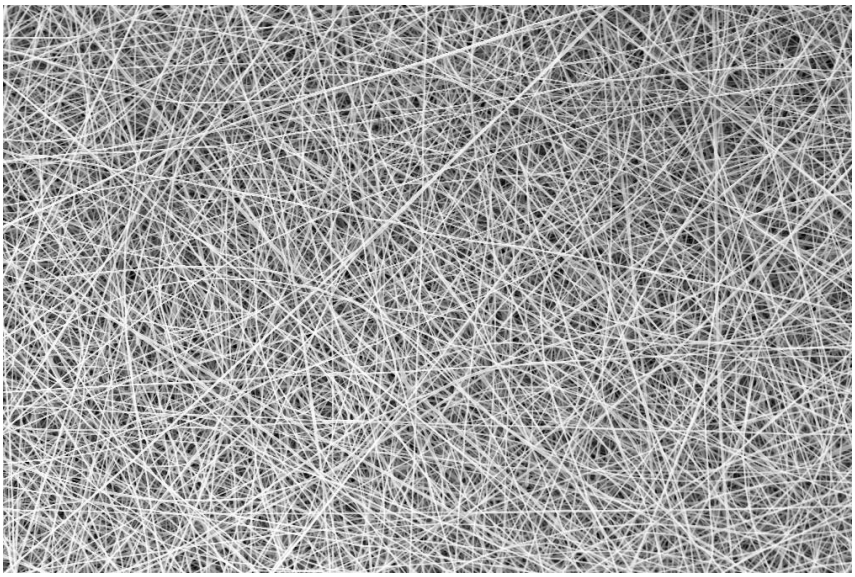
E-Spin Nanotech manufactures advanced electrospun PVA nanofiber membranes designed for a wide range of filtration and functional material applications. Polyvinyl alcohol (PVA) is a water-soluble synthetic polymer known for its excellent film-forming ability, high hydrophilicity, good mechanical properties, and biocompatibility.

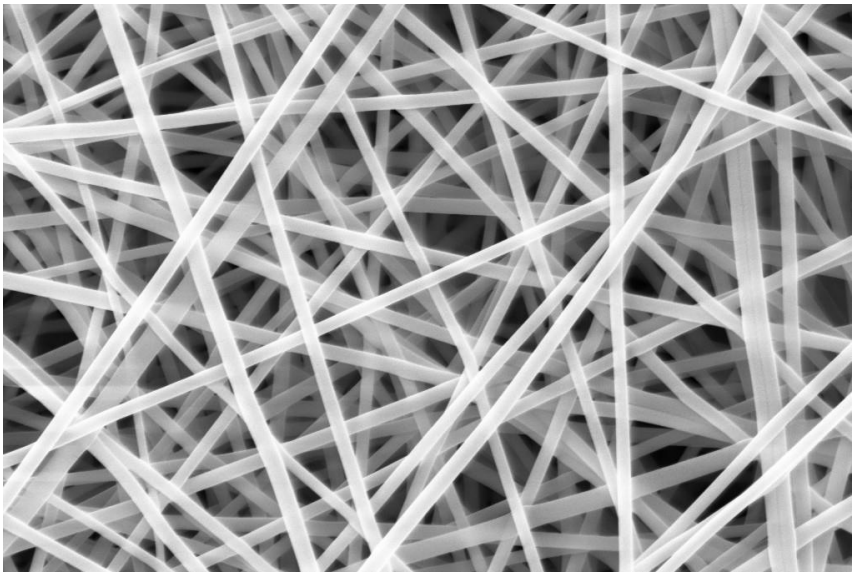
Electrospun PVA nanofiber membranes exhibit high porosity, interconnected pore structure, and ultrafine fibre diameter, which provide large surface area and efficient transport properties. These characteristics make them highly suitable for drug delivery, biomedical materials, and functional coatings.

Due to their hydrophilic nature and chemical versatility, PVA nanofibers can also be easily modified or crosslinked to enhance their stability and performance for specific applications such as adsorption, separation processes, and environmental remediation.

Key Applications

- Drug delivery
- Biomedical and healthcare materials
- Functional coatings
- Environmental remediation and adsorption systems





Nanofiber Spinning Capability

- Pure PVA
- Blended with other polymers like chitosan, gelatine, nanoparticles, additives etc.

Membrane Construction

- Free-standing nanofiber membrane
- Supported on substrates (e.g., aluminium foil, non-woven PP fabric, etc.)

Nanofiber Diameter Range

- 100-600 nm

Thickness and weight Density

- Thickness 50-100 µm
- Up to 8.0 gsm

Custom Sizes

- Rolls
- Sheets (10 × 10 cm, 14 × 28 cm, 25 × 200 cm)
- Custom-sized patches (as per specific requirements)